



Impact of Governance Structure on the Effective Implementation of the European Plastic Circular Economy—The Interlinkage of Law, Standards, and Industrial Practice

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Abstract: This study explored the governance structure that supported the implementation of the European plastic circular economy by examining the relationships among legal requirements, technical standards, industrial practices, and digital infrastructure. In recent years, the European Union (EU) has strengthened its circular economy policies through measures such as the Packaging and Packaging Waste Regulation (PPWR), the Ecodesign for Sustainable Products Regulation (ESPR), and the Digital Product Passport (DPP). Researchers initially analyzed the impact of environmental regulations on global markets, global value chain (GVC) governance, and digital traceability systems as separate topics. Relatively little attention has been paid to how these elements incorporated as a single governance system. To bridge this research gap, this study introduced the concept of Institutional Translation and proposed an analytical framework encompassing three institutional domains, i.e., law, standardization, and industrial practice, to be supported by a cross-cutting digital infrastructure. Using qualitative mechanism mapping, the study explored how policy objectives were translated into operational practices. The findings underscored the effectiveness of the EU circular economy framework though it depended not simply on regulation, standards, or digital systems in isolation. The way these elements interacted was indeed the critical impetus. Policy objectives were translated into compulsory market conditions through legal mandates, operationalized through standards developed in interaction with the industry, and supported by digital infrastructures that facilitated traceability and compliance verification across value chains. In summary, the European circular economy framework extended beyond conventional environmental regulation by combining legal requirements, standardization processes, industrial practices, and digital infrastructures in a coherent governance arrangement. By introducing the concept of Institutional Translation, this study explained how circular economy policies were put into practice across complex value chains.

Keywords: Circular economy; Plastic circularity; Governance structure; Institutional translation; Standardization; Digital product passport; Global value chains

1 Background

In Europe, the transition toward a circular economy along with a decarbonized society has been prioritized by policymakers. To support this transition, the European Union (EU) has introduced a series of regulatory initiatives aimed at increasing the utilization of recycled materials and improving product circularity. These include the Packaging and Packaging Waste Regulation (PPWR), which establishes recycled-content and recyclability requirements for packaging; the Ecodesign for Sustainable Products Regulation (ESPR), which introduces sustainability requirements for products; and the Digital Product Passport (DPP), which targets at improving the availability and traceability of product information across value chains. These policies are intended not only to reduce environmental impacts, but also to promote the development of circular economy-related industry including the secondary raw material markets and the wider adoption of circular practices throughout value chains.

Under the conventional perspective, environmental policy has primarily been regarded as a means of controlling environmentally harmful activities and addressing market externalities, such as environmental costs that are not

reflected in market prices (e.g., pollution and waste generation). More recently, however, policy interventions have gradually sought to reshape markets and production systems in ways that validate the transition toward a circular economy. To guarantee a smooth transition, there should be a wide range of institutional and organizational arrangements, including regulatory frameworks, industrial coordination, market incentives, information sharing, and technological innovation.

Among these elements, the interaction among legal frameworks, standardization activities, and digital information infrastructures has become particularly crucial for translating policy objectives into operational practices. For this purpose, these mechanisms are being developed in parallel, yet their interdependence remains only partially revealed.

2 Literature Review

2.1 Environmental Regulation and Market Formation

Environmental regulation has traditionally been understood as a mechanism for correcting externalities and reducing environmental burdens. Another line of research, however, suggests that regulation could shape markets and stimulate technological change. Rather than simply imposing additional costs on firms, well-designed environmental policies may encourage innovation and boost competitiveness. This perspective, commonly known as the Porter Hypothesis, provides a theoretical basis for treating regulation as a driver of industrial transformation [1]. Recent conceptualizations further emphasize that systemic policy interventions are required to shift market dynamics from linear to circular models [2]. This paradigm shift demands a comprehensive governance framework that moves beyond traditional end-of-pipe waste management and instead promotes holistic, system-wide transformations encompassing reduced consumption and high-quality material circulation [3].

Subsequent studies further developed this perspective by showing that policy instruments, such as carbon pricing, could point out the direction of technological advancement and investment [4]. Research on mission-oriented innovation policy likewise emphasized the role of public intervention in creating and shaping new markets, particularly in sustainability transitions [5]. In this context, future-oriented policy mixes that combine market-pull and technology-push instruments have been identified as critical enablers for long-term industrial restructuring [6].

Together, these studies suggested that environmental policy might contribute to the formation of markets for circular products and materials. Empirical evidence indicates that the orientations and strategic positions of diverse industrial actors heavily influence how these regulatory frameworks are customized in practice [7]. This implies that policy design is not a one-way process; rather, it is shaped by negotiations and power dynamics among stakeholders. Incumbent firms often leverage their market dominance to align rules with their existing business models, whereas emerging, sustainability-oriented pioneers push for more stringent standards to secure a competitive advantage. Consequently, the final regulatory outcomes reflect the strategic compromises made by these conflicting industrial forces [7, 8]. At the same time, they provide relatively limited explanation of how these policy objectives become embedded in the day-to-day operation of complex industrial systems. This highlights a multi-level governance gap between macro-level political discourse and micro-level industrial execution [9].

2.2 Standards and Governance in Global Value Chains

The smooth coordination of secondary raw materials across firms from multiple EU countries has been a critical issue in the promotion of circular economy. Such coordination have been discussed in the literature on global value chains (GVCs). A widely adopted framework helps explain governance structures through three factors: The complexity of transactions, the extent of information to be codified, and the capabilities of suppliers [10]. The complexity of transactions indicates how detailed and non-standardized information is processed among firms and suppliers across the GVCs, as can be seen in the automobiles and electronics sectors. Information codification refers to the extent to which complex product specifications could be translated into structured manuals or technical standards, such as those implemented by the Japanese bicycle company, Shimano. The capabilities of suppliers measure whether the suppliers, such as local factories, possess the internal competence and technology to meet the lead firms' strict demands. Within the clearly defined framework, standards are emphasized as they allow product specifications and process requirements to be communicated and applied across organizations.

Later, more studies expanded this view by validating that standards and certification schemes performed more than merely facilitating transactions. They also influenced market access, defined acceptable practices, and shaped broader institutional arrangements within value chains [11]. Structural frameworks for circular value chains increasingly demonstrate that relational governance is essential for managing the continuous circulation of secondary resources across firm boundaries. Research on standardization has demonstrated that standards do not emerge solely from technical rationality, but are shaped through interactions among firms, industry associations, and standard-setting organizations that negotiate and institutionalize technical knowledge into widely accepted rules [12, 13]. These studies, however, did not explicitly integrate the governance perspective of GVCs, particularly regarding the interaction among legal requirements, standards, and inter-firm coordination. Particularly in the plastics sector, technical standardization acts as a critical mechanism to mitigate material quality risks and streamline processing criteria among fragmented

supply chain tiers [14], thereby facilitating coordination among diverse actors engaged in circular value chains. Such coordination is increasingly recognized as essential for enabling circular economy practices across GVCs. Nevertheless, how regulatory mandates interact with GVC governance mechanisms to shape such coordination remains insufficiently understood, representing a key frontier for empirical investigation [15].

2.3 Digital Infrastructure and Information Governance

Discussions to date on the circular economy have highlighted the importance of information management across product life cycles. Among the proposed instruments, DPPs have attracted attention as tools for improving transparency and traceability in the circular economy [16].

Recent reviews attempted to determine the technical architecture of DPP systems, including issues of interoperability, data exchange, and challenges of implementation [17, 18]. For instance, if Manufacturer A defines a material as “PP-Polypropylene” and Recycler B’s database recognizes it only as “Resin Code 5”, the system breaks down. Developing unified common data dictionaries (CDDs) is therefore critical to identically interpret data in GVC. Advanced data architectures for DPP systems highlight that such standardized semantic structures are crucial for overcoming cross-border technical barriers [19]. Other studies emphasized that DPPs could also function as governance mechanisms by structuring information flows and reducing information asymmetries among actors involved in circular value chains [20]. In plastic value chains specifically, digital traceability tools reduce information asymmetry by enabling the exchange of verified information on material composition, recycled content, and product provenance across upstream manufacturers and downstream recyclers, thereby facilitating higher-quality plastics recycling [21]. An information gap existed in many cases involving both upstream manufacturers and downstream recyclers; to deal with the issue, DPPs could act as an organizational governance mechanism that defined the owner and the content of the shared data [22].

While this body of research demonstrated the potential of digital infrastructures to support practices of the circular economy, their relationship with legal requirements and standardization activities remained only partially explored. A comprehensive information governance framework is required to bridge the gap between regulatory transparency mandates and operational standard-setting processes.

Together, previous studies generated important insights into environmental regulation, standardization, and digital traceability systems. These elements, however, have largely been understood as discrete entities. Studies on environmental regulation focused on market creation and innovation, while GVC research explored coordination through standards and DPP research focused on information technology and infrastructure.

This limitation becomes more apparent in the European circular economy, where policy measures, standardization efforts, and digital initiatives are evolving alongside one another. In particular, the processes through which policy objectives are translated into standards, operational practices, and digital verification mechanisms have not been systematically reviewed. Interpreting how these elements interact within a common analytical framework is essential for examining the actual execution of circular economy policies.

3 Research Objectives

The agenda of the European circular economy provides an opportunity to study how different institutional arrangements collaborate in practice. Legal obligations, technical standards, industrial initiatives, and digital information systems are expected to contribute to the transition toward circularity, yet the relationships among these elements remain only partly understood.

With a particular focus on the EU plastics sector, this study examined how the objectives of circular economy policies were translated into operational practices through the interaction of legal requirements, technical standards, industrial practices, and digital infrastructure.

4 Analytical Framework and Methodology

4.1 Analytical Framework

European standards (EN) and draft European standards (prEN) are used in this study to refer to formal European Norms and their draft versions, respectively. The concept of “Institutional Translation” provides the analytical foundation of this study. Institutional Translation refers to the sequential process through which high-level policy objectives are progressively converted into concrete and enforceable market conditions as they pass through successive institutional domains.

As illustrated in Figure 1, the analytical framework focuses on three institutional domains, including law, standardization, and industrial practice, while digital infrastructure is considered a cross-cutting element. In the context of EU plastic circularity policy, a political objective such as “increasing plastic recycling and utilization of recycled plastics” is translated across the following domains:

The legal domain converts this objective into binding legal mandates such as mandatory recycled content targets under the PPWR that define the conditions for market participation. The key inquiry here is the transition from discretionary guidelines to mandatory requirements of market participation.

The standardization domain then translates these mandates into objective, measurable, and verifiable technical criteria such as the Design for Recycling (DfR) guidelines in the prEN 18120 series. The effectiveness of this translation depends on whether the resulting standards are both technically sound and practically implementable, as they developed through interactions between public authorities and industry actors.

The industrial domain operationalizes these criteria into concrete business practices, such as the adoption of DfR in product development and shifts in procurement strategies toward secondary raw materials often finalized through formal declarations of conformity. These domains are supported by cross-cutting digital infrastructure. Rather than constituting a separate sequential stage, digital tools such as the DPP help uphold information sharing, secure a traceability, and verify compliance across the three domains.

The central questions raised in this study concerned how these institutional domains interacted in the implementation of circular economy policies and whether their interactions could support effective policy implementation.

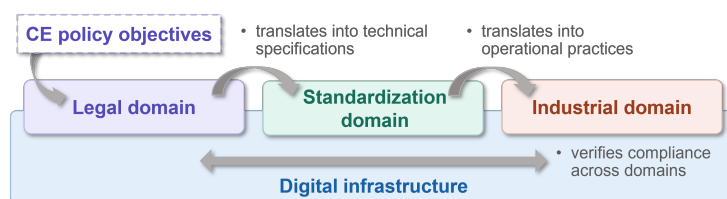


Figure 1. Analytical framework of institutional translation

Note: CE = circular economy.

4.2 Methodology: Institutional Translation Analysis

The study adopted a qualitative methodology based on documentary analysis of EU regulatory instruments, standardization documents, and industry guidelines. The analysis traced how selected policy instruments were translated across the three institutional domains within the analytical framework of institutional translation, following two documented pathways.

The first was a bottom-up pathway, in which industry-led practices and standards were progressively formalized through European standardization processes and subsequently referenced in legal instruments. The second was a top-down pathway, in which administrative mandates drove standardization with the industry, to provide input of implementability. These two pathways were not mutually exclusive; rather, their interaction reflected the hybrid character of EU governance under study. The analysis validated whether output from one domain were incorporated into subsequent domains and how these interactions contributed to policy implementation.

The instruments selected PPWR, ESPR, DPP, EN 15343, and the prEN 18120 series, representing one of the most institutionally documented cases within the EU plastic circularity policy, to enable systematic analysis of institutional translation across domains. As this study drew on multiple regulatory and industry cases rather than a single product-level case study, the findings were indeed a conceptual synthesis.

5 Institutional Translation in Practice

5.1 Legal Domain: Mandatory Market Creation and Regulatory Stabilization

Several structural conditions affect the functioning of a circular economy. Currently, recycled material markets appear fragmented across the EU states, and recycled plastics often fail to circulate as resources, due to restrictive legal classifications under waste regulations.

The PPWR, which replaces the previous Packaging and Packaging Waste Directive (PPWD), serves as the primary legal driver for creating a mandatory market for circular plastics in Europe. Unlike the previous directive, the PPWR establishes direct obligations for firms and manufacturers operating under the EU regulatory framework (referred to in regulatory texts as “economic operators”). It defines the precise conditions under which packaging could access the EU internal market. This regulatory shift is characterized by the following developments of the plastic circular economy.

The PPWR introduces a rigorous grading system for DfR. Under Article 6, all packaging must be recyclable by 2030, and its recyclability will be evaluated on a scale from A to C [23]. The regulation establishes a strict timeline for market access; from 2038 onwards, packaging that fails to meet a minimum recyclability grade (e.g., below grade B) may be prohibited from being placed on the market. This grading mechanism is intended to support “law-driven market creation” by turning recyclability into a mandatory condition for market access.

The legal domain is further strengthened by two key policy instruments that provide the institutional infrastructure for a circular market:

The first is the End-of-Waste (EoW) Criteria. EoW criteria have been discussed as a policy instrument for reducing regulatory fragmentation among Member States by establishing common requirements under which recycled materials cease to be legally regarded as waste and become secondary raw materials [24]. Once adopted, these criteria would legally define the state at which recycled plastic ceases to be “waste” and becomes a “secondary raw material”. This is expected to simplify administrative procedures for recyclers and facilitate the free movement of quality-defined recyclates within the Single Market.

The second is the Circular Economy Act, announced by the Commission in 2025, which aims to provide a holistic framework for the “Single Market for secondary raw materials”, potentially expanding the governance model beyond specific sectors like packaging to the entire industrial economy [25].

Through these interconnected legal instruments, PPWR for product-specific design requirements, customs, and monitoring for fair competition, and EoW for material status, the EU strives to build a legal foundation that supports the transition to a circular economy for plastics.

5.2 Standardization Domain

If law-driven market creation is to function effectively, abstract legal mandates, such as the requirements for recyclability or mandatory recycled content, should be translated into technical specifications that are objective, measurable, and verifiable. At the standardization domain, harmonized standards offer the technical means to demonstrate conformity with these legal requirements. In the context of the PPWR, this translation occurs through a combination of industry-led initiatives and regulatory-driven standardization.

5.2.1 Securing implementability through industry-led standards (bottom-up approach)

To ensure that technical criteria are grounded in industrial reality, the European standardization processes often incorporate empirical evidence from the private sector through a “bottom-up” process. This involves formalizing industry-developed guidelines, which are initially established as de facto “Forum standards”, into public EN standards after their social implementability has been proven in the market [26].

One illustration of this translation process is the development of the prEN 18120 series, a set of draft standards intended to assign comprehensive DfR guidelines for various plastic packaging categories [27]. The European Committee for Standardization (CEN) has acted as a platform to codify validated technical protocols from sector-specific organizations, ensuring that the resulting standards reflect the current technical capabilities of the recycling infrastructure.

The development of prEN 18120-7:2024 (polyethylene (PE) and polypropylene (PP) flexible packaging) illustrates how industry guidance and formal standardization can evolve together [26, 27]. A comparison between the RecyClass DfR methodology and the draft prEN 18120-7 framework reveals several common technical criteria. RecyClass, an industry-led initiative that provides DfR guidelines and recyclability assessment methodologies for plastic packaging, has established criteria related to the main polymer composition (for example, the threshold of more than 95% PP for the highest recyclability grade), the evaluation of near-infrared (NIR)-detectable colours, and the treatment of non-polyolefin components such as labels and closures [26, 27]. Similar correspondences can also be observed in the treatment of printing inks, functional layers, and copolymers, which are addressed in both the RecyClass guidance and the draft EN standard [26, 27]. CEFLEX, a collaborative initiative to build a circular economy for flexible packaging, has developed Designing for a Circular Economy (D4ACE) guidelines based on extensive testing and stakeholder collaboration across the flexible packaging value chain. These guidelines provide operational criteria for recyclability assessment and packaging design, and illustrate how industry-generated technical knowledge can support subsequent standardization activities and regulatory implementation [27].

Similarly, for the Polyethylene Terephthalate (PET) sector, the European PET Bottle Platform (EPBP) has developed detailed DfR guidelines addressing the compatibility of labels, adhesives, closures, barrier layers, and other packaging components within bottle-to-bottle recycling systems [28]. These guidelines have contributed to the development of harmonized recyclability requirements across the European PET value chain.

These examples illustrate how industry-led recyclability guidelines and assessment methodologies are being translated into formal EN through the prEN 18120 series [29]. This process enables recycling principles to be institutionalized within the mandatory framework, while ensuring that its underlying technical specifications remain closely connected to the empirical data accumulated by industry forums. By translating these detailed industrial requirements into a common regulatory language, the EU ensures that legal mandates are not merely ambitious policy goals but are “operationally implementable” technical requirements supported by the entire value chain.

5.2.2 Accelerating transformation through policy-led mandates (top-down approach)

While bottom-up standards ensure practical feasibility, the EC often utilizes a “top-down” approach to force rapid industrial transformation when voluntary initiatives are insufficient to meet environmental targets. Under the

European Green Deal, administrative mandates directly dictate technical specifications through formal Standardization Requests for the implementation of the PPWR and ESPR [30].

The policy-led shift can be seen clearly in the regulation mandating the use of Post-Consumer Recycled (PCR) content. Article 7 of the PPWR sets minimum recycled-content targets for selected products, including a requirement that PET beverage bottles contain at least 30% recycled material by 2030 [23]. Meeting these targets requires harmonized methods for calculating, verifying and documenting recycled content. Existing standards such as EN 15343 provide a technical basis for plastics traceability and recycled-content calculation, while industry certification schemes such as RecyClass operationalize these requirements through auditable supply-chain verification systems. As these standards and certification mechanisms become incorporated into conformity assessment procedures that support the Declaration of Conformity (DoC) framework under the PPWR, policy objectives are translated into concrete obligations for economic operators.

The interaction between bottom-up industrial practices and top-down regulatory measures creates links across institutional domains, helping to stabilize the emerging market for circular plastics. The legal requirements of recyclability and recycled content are translated into measurable indicators, while operationally and technically validated industry standards ensure that these indicators are feasible in practice. Once incorporated into EN standards and referenced in secondary legislation (specifically delegated and implementing acts), compliance with these specifications becomes the responsibility of economic operators, linking legal obligations to industrial practices. Consequently, the European standardization process functions as a bridge between policy objectives and market implementation.

In this way, the standardization domain reflects a hybrid governance pattern. Administrative action provides uniformity and ambition across the Single Market, while industry-led validation guarantees practical implementability. The iterative process from industry standards to harmonized EN standards ensures that legal mandates are grounded in industrial reality, producing an institutional structure that effectively translates regulatory objectives into market-ready technical requirements.

5.3 Industrial Domain: Reorganizing Industrial Practices and Securing Supply Chain Compliance

The industrial domain is another stage of institutional translation, where abstract legal mandates and technical specifications are operationalized into concrete business practices. Under the evolving European regulatory framework such as the PPWR and ESPR, economic operators are required to move beyond incremental environmental improvements and fundamentally reorganize their core operations.

This reorganization is first manifested in the integration of circularity requirements at the earliest stages of the product lifecycle. In terms of product design, firms are adopting the DfR criteria and grading schemes (e.g., A to C; to be applied from August 2026 under the PPWR Article 6 [23]) defined in the standardization domain to improve recyclability and ensure compatibility with existing collection and sorting infrastructures. At the same time, the recycled content requirements under the PPWR (Article 6) are reshaping procurement strategies by requiring manufacturers to secure reliable supplies of secondary raw materials [23]. By formalizing demand that was previously voluntary or fragmented, these requirements support the development of a more predictable market for recyclates.

The mechanism connecting these industrial practices to legal responsibility is the EU DoC. Under the PPWR and ESPR, manufacturers are legally responsible for ensuring that products satisfy the applicable technical requirements before they are placed on the market. Compliance is ultimately confirmed through a formal conformity assessment, supported by technical documentation including test reports prepared in accordance with harmonized EN standards. These serve as objective evidence which have already fulfilled those requirements. By signing the DoC, manufacturers convert technical implementation into a legally binding declaration, allowing products to circulate within the Single Market while also providing a basis for market surveillance activities.

Some firms have also participated in industry initiatives such as RecyClass and CEFLEX, where practical methods and evaluation protocols have been developed before the corresponding legal requirements come fully into force. When these industry-based approaches are later incorporated into formal EN standards, participating firms already possess much of the technical capability and supply-chain coordination needed for compliance. The industrial domain therefore serves as a place where emerging standards and regulatory expectations are tested through actual business practice, making it possible to assess whether they can work under real operating conditions. As a result, legal requirements are more likely to be implemented in ways that fit the realities of industrial value chains.

5.4 Digital Domain: Constructing an Operational Infrastructure

The digital domain provides the information infrastructure necessary for connecting legal requirements and technical standards to the physical reality of products across complex value chains. While legal mandates establish regulatory objectives and technical standards specify methods for demonstrating compliance, circular economy governance remains limited unless the resulting information can be retained, shared, and assessed across different

actors. In this sense, the digital domain functions as the operational infrastructure that links governance requirements to material flows.

The central instrument within this digital infrastructure is the DPP, introduced under the ESPR and expected to be progressively applied across product groups [31]. The DPP serves as a digital record that stores and communicates product sustainability information, including material composition, recycled content, carbon footprint, and relevant conformity documentation, throughout the product lifecycle.

First, the DPP establishes traceability by linking physical products with digital records throughout the value chain. Product identifiers, such as Quick Response (QR) codes, digital watermarks, or other machine-readable carriers, allow information on material composition, recycled content, and conformity documentation to remain associated with products as they move through manufacturing, use, collection, and recycling stages. This enables information generated during production, including evidence documented through mechanisms such as the EU DoC, to remain accessible to downstream actors, including recyclers and market surveillance authorities.

However, the transmission of information alone is insufficient if different actors interpret the same data differently. As recognized in the European Interoperability Framework, semantic interoperability requires that the meaning of exchanged information be preserved across different systems and actors, ensuring that “what is sent is what is understood” [32]. A prerequisite for the DPP to function is therefore semantic interoperability, in other words, the condition where product information retains a common meaning as it moves across manufacturing, collection, recycling, and regulatory systems. To prevent data fragmentation, the EU is prioritizing the development of CDDs and ontologies to ensure that terms and attributes are defined consistently across the Single Market. This semantic alignment is particularly vital for bridging the information gap between the “artery” (manufacturing) and “vein” (recycling) sectors. Emerging standards such as the prEN 18216 series further contribute to establishing common frameworks for digital data exchange. Emerging initiatives for DPP-related standardization are also expected to contribute to common frameworks for digital data exchange. Such efforts aim to improve the consistency with which information generated by manufacturers can be interpreted and applied by recyclers, regulators, and other actors across the value chain.

Finally, the digital domain provides the basis for verification by connecting legal obligations, technical standards, and conformity evidence within a common digital framework. Compliance information generated through harmonized standards and documented in instruments such as the EU DoC could, in principle, be associated with individual products and assessed according to common criteria across the Single Market. Emerging DPP-related standardization activities are expected to strengthen market surveillance and facilitate the verification of requirements, including mandatory recycled-content obligations under the PPWR. In this way, the digital domain functions not merely as an information repository, but as the infrastructure through which evidence generated by legal, technical, and industrial processes can be retained, shared, and verified across the value chain.

6 Discussion

6.1 Synthesis: The Mechanism of Institutional Translation

The analysis of the European plastic circular economy indicates that market formation depends not on legal mandates alone, but on the way policy instruments are connected across the four domains of governance.

Figure 2 illustrates this process as a chain of institutional translation. The legal domain (left) creates mandatory demand through specific regulatory requirements such as the PPWR Article 7 target of 30% PCR content for PET bottles by 2030, and the Article 6 recyclability grading system from A to C. These requirements do not operate directly on industrial practice; instead, they are first translated into technical specifications within the standardization domain (center). This translation takes two forms: a top-down pathway, in which regulatory requirements such as the PPWR recycled-content targets create demand for harmonized methods of calculation, verification, and conformity assessment, supported by standards such as EN 15343, and a bottom-up pathway, in which industry-validated protocols, developed through forums such as RecyClass and CEFLEX, are formalized into draft standards such as prEN 18120. The industrial domain (right) then translates these criteria into business practices, operationalized through DfR grading criteria and the DoC, which links technical compliance to legal market access. Underlying all three domains, the digital infrastructure provides the operational protocols that enable compliance verification and traceability across the value chain.

The effectiveness of the EU circular economy framework emerges from the mutual reinforcement among political directionality, legal compulsion, industrial implementability, and digital verification. In this governance structure, policy objectives are translated into mandatory market conditions through legal instruments, operationalized through technically and operationally implementable standards, and stabilized through digital infrastructures that ensure transparency, traceability, and verifiability across value chains.

It was also revealed that standardization and data modeling, if decoupled from the legal framework, remain insufficient to drive industrial transformation. Without such cross-domain connections, technical specifications often fail to overcome market failures arising from information asymmetry and high transaction costs. The effectiveness of

the EU model lies in its ability to transform voluntary standards into de facto mandatory requirements by referencing them in secondary legislation and linking them to the EU DoC. This ensures that technical criteria function as defined conditions for market participation.

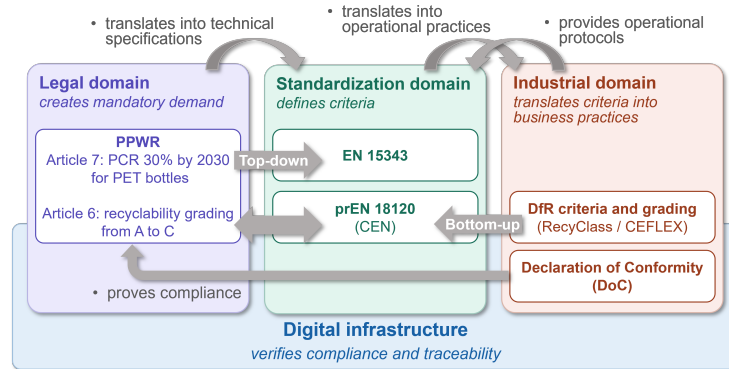


Figure 2. Mechanism map of institutional translation

Note: PPWR = Packaging and Packaging Waste Regulation; PCR = Post-Consumer Recycled (contents); PET = Polyethylene terephthalate; EN = European standard; prEN = draft European standard; DfR = Design for Recycling; RecyClass = an industry-led initiative that provides DfR guidelines and recyclability assessment methodologies for plastic packaging; CEFLEX = a collaborative initiative to build a circular economy for flexible packaging.

6.2 Hybrid Governance Pattern in Standardization

The standardization process reflects a hybrid governance pattern, in which top-down administrative direction and bottom-up industry input operate in combination. While industry-led “Forum standards” (e.g., RecyClass) provide the necessary empirical evidence and “social implementability”, administrative leadership through the EC ensures that these standards are unified across the Single Market to prevent market fragmentation. This combination reduces the risk that firms selectively apply only the least demanding indicator, and provides a degree of regulatory consistency that supports longer-term planning by industry.

6.3 Role of Digital Infrastructure in Governance

The digital domain plays an enabling role within the governance structure. The DPP and semantic interoperability are not ends in themselves, instead they provide the informational conditions under which legal requirements and technical standards can operate across complex value chains. The capacity for information to be consistently interpreted across different systems, what might be described as “what is sent is what is understood”, is a functional prerequisite for traceability and compliance verification.

Where digital infrastructure is absent or inconsistent, there is limited ability to verify compliance at the product level, i.e., across model, batch, and item granularities. In this sense, digital systems may contribute to the effectiveness of legal and technical requirements, by providing the means to connect regulatory obligations to physical product flows. The extent to which this potential is realized in practice will depend on the adoption and interoperability of emerging standards such as the prEN 18216–18223 series.

6.4 Implications for Other Regions and Sectors

The European circular economy framework combines regulatory, technical, and digital instruments in a way that bypasses conventional environmental regulation. The structure analyzed in this study connected legal requirements, standardization, conformity assessment, and digital verification to support a shared policy objective. This reflects an approach to governance in which multiple instruments are coordinated rather than operated independently.

The transition toward Regulation over Directives, combined with harmonized EoW criteria, is intended to create a level playing field that facilitates cross-border circulation of resources. These developments also matter for countries and firms outside the EU, thus affecting not only access to the European market but also broader approaches to governance. Limited participation in the development of international rules, especially semantic data models and grading standards, may render it more difficult for firms to satisfy future EU requirements. Meanwhile, the analytical framework developed in this study may provide a basis for assessing circular economy governance in other regional contexts.

7 Conclusions

This study investigated the institutional structure through which circular economy policy for plastics was operationalized in Europe, in accordance with the concept of institutional translation as an analytical lens. A central observation of this study was that circularity took shape through the interaction of three institutional domains: law, standards, and industrial practice, and their connections to a shared digital infrastructure. The effectiveness of this arrangement lied in the links among these domains, through which policy objectives were translated into operational practice.

This study incorporated insights from research on environmental regulation, standardization, and digital information governance, as well as insights gained from actual policy processes related to secondary plastic materials. The four interrelated domains encompassing legal, standardization, industrial, and digital converged to help examine how governance structures supported the implementation of circular economy policies. The effectiveness of the policy depended not on any single instrument, but on the degree of coordination among the instruments that constituted the broader governance structure.

Another implication of this analysis was that the European circular economy framework operated differently from conventional environmental regulation. Rather than relying on regulation alone, the EU approach combined legislation, standardization, conformity assessment, and digital verification to carry out policy objectives within the value chain, while simultaneously fostering the development of related industries.

The analysis presented here interrogated the institutional relationships among legal requirements, technical standards, industrial practices, and digital verification in the EU plastics sector, using the PPWR-related standards as an illustrative example. The study does not draw on empirical evidence from interviews with industry associations, standardization bodies, or firms directly involved in developing and implementing these mechanisms. Further research could explore how these actors contribute to the process of institutional translation and how such institutional linkages are established, negotiated, and sustained over time.

While this study drew reference from the case of plastic packaging, more insights could be obtained from scrutinizing the extent to which the same pattern of institutional translation occurred in other sectors such as batteries, automotive products, textiles, and electronics. Comparative analyses across various sectors could help assess the conditions under which the proposed analytical framework may be applicable. Such work could clarify how circular economy policies take shape across different institutional settings. It would also be advantageous to examine whether actors throughout the value chain, from small recyclers to downstream manufacturers and retailers, have the capacity to engage with the digital systems which are progressively utilized. The compliance of technical standards with regulatory requirements would enlighten stakeholders with relevant insights for further improvement.

Author Contributions

Conceptualization, I.K. and S.H.; methodology, I.K.; validation, I.K. and S.H.; formal analysis, I.K.; investigation, I.K. and N.A.; writing—original draft preparation, I.K.; writing—review and editing, S.H. and N.A.; visualization, I.K.; project administration, I.K.; funding acquisition, S.H. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data supporting the findings of this study are derived from publicly available documents and sources cited in the References. No new datasets were generated or analyzed during the current study.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the authors used AI-assisted writing tools for language editing and proofreading purposes. All intellectual content, analytical framework, and conclusions are the authors' own work. The authors reviewed and edited all AI-assisted content and took full responsibility for the accuracy and integrity of the manuscript.

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